

# WC 2026 McHale–Copula Backtest Report

Moneyline Value Betting & Asymmetric-Result Hypothesis

WC Prediction Project

July 2026

## 1 Setup

**Model.** We use a McHale–Copula model: negative-binomial goal marginals linked by Frank’s copula, with the copula intensity parameter  $k$  held constant. Optimal hyperparameters ( $\alpha = 50.0$ , `k_mode = const`) were selected on the validation set. For each fixture the model produces *neutral* win/draw/loss probabilities by averaging predictions under both home/away orientations.

**Training window.** The model is trained on all available data strictly before the test set ( $< 2025-10-01$ ), combining the original training split and the validation split to give 3,148 matches. All WC 2026 matches (group stage + Round of 32, June 11 – July 2) are out-of-sample test observations.

**Market data.** Match-level prediction-market odds are sourced from Kalshi (finalized contracts):

- **KXWCGAME:** regulation-time moneyline (win / draw / loss), 73 fixtures, 219 contracts.
- **KXWCTOTAL:** total-match-goals over/under lines (0.5–6.5+), 82 fixtures, 544 contracts.

**PnL accounting.** All positions are \$1 notional per contract (prediction-market convention):

$$\text{PnL}_{\text{BUY}} = \mathbf{1}[\text{hit}] - p, \quad \text{PnL}_{\text{SELL}} = p - \mathbf{1}[\text{hit}],$$

where  $p$  is the market implied probability at the time of the snapshot.

## 2 Backtest 1: Moneyline Value Betting

### 2.1 Strategy

For every WC 2026 moneyline contract, we compare the model’s neutral WDL probability  $\hat{q}$  against the market implied probability  $p$ :

$$\begin{aligned} &\text{BUY if } \hat{q} > p + 0.02 \quad (\text{market undervalues outcome}), \\ &\text{SELL if } p > \hat{q} + 0.02 \quad (\text{market overvalues outcome}). \end{aligned}$$

The 2-cent threshold filters out noise near the bid–ask spread.

Table 1: Overall moneyline backtest results (72 matches, all WC 2026).

| Subset          | Bets | Staked | Net PnL | ROI   |
|-----------------|------|--------|---------|-------|
| All bets        | 182  | \$182  | \$-4.15 | -2.3% |
| BUY             | 100  | \$100  | \$-3.28 | -3.3% |
| SELL            | 82   | \$82   | \$-0.87 | -1.1% |
| By outcome type |      |        |         |       |
| Home win        | 60   | \$60   | \$-2.02 | -3.4% |
| Draw            | 56   | \$56   | \$-1.32 | -2.4% |
| Away win        | 66   | \$66   | \$-0.81 | -1.2% |

Table 2: Bucket analysis – moneyline backtest.

| Bucket                                    | Bets | Net PnL | ROI    | Hit% |
|-------------------------------------------|------|---------|--------|------|
| Both teams score $\geq 4$ goals           | 0    | –       | –      | –    |
| Asymmetric (min $\leq 1$ , max $\geq 3$ ) | 70   | \$-7.97 | -11.4% | –    |
| BUY contracts                             | 44   | \$-4.34 | -9.9%  | 2%   |
| SELL contracts                            | 26   | \$-3.63 | -14.0% | 88%  |
| SELL with implied $> 80\%$                | 17   | \$+0.96 | +5.6%  | 82%  |
| BUY with implied $> 80\%$                 | 0    | –       | –      | –    |

## 2.2 Results

## 2.3 Discussion

The overall  $-2.3\%$  ROI is consistent with the Kalshi round-trip vig ( $\approx 2-4\%$ ), suggesting the McHale–Copula model produces predictions close to market-efficiency but adds no systematic edge.

The only profitable pocket is **selling heavily-implied favourites** (market implied  $> 80\%$ ): 17 contracts,  $+\$0.96$ ,  $+5.6\%$  ROI. Although the favourite still won 82% of the time, the market slightly over-prices certainty in small-volume WC prediction markets, leaving a marginal sell edge.

Asymmetric matches (one team scores 3+, other scores  $\leq 1$ ) were harmful: the model’s BUY signals in lopsided matches had only a 2% hit rate, and SELL signals were wiped out when draws unexpectedly occurred (e.g. Portugal 1–1 DR Congo, Canada 1–1 Bosnia). The model systematically over-estimates draw probability for mismatched pairs.

## 3 Backtest 2: Asymmetric-Result / Total-Goals Hypothesis

### 3.1 Hypothesis

When the prediction market prices one team as a heavy favourite (implied win probability  $> 65\%$ ), the match is more likely to produce an asymmetric result: the strong team scores 3+ goals while the weak team scores 0–1. This should push total match goals above what the over/under market prices, creating a potential BUY edge on higher lines.

### 3.2 Strategy

1. Identify matches where  $\max(p_{\text{home}}, p_{\text{away}}) > 0.65$ .

2. On those 32 matches, BUY over 2.5, 3.5, 4.5, 5.5, and 6.5 total goals (*5 contracts per match, \$160 total staked for 32*

*Note on “under 0.5 for the weaker team”:* Kalshi offered only total-match-goals markets, not individual team scoring markets. “Team B scores 0 goals” cannot be directly bet on with the available data.

### 3.3 Results

Table 3: Totals backtest by over line (32 qualifying matches).

| Over line    | Bets       | Hits      | Hit%         | Avg implied | Net PnL        | ROI          |
|--------------|------------|-----------|--------------|-------------|----------------|--------------|
| 2.5          | 32         | 22        | 68.8%        | 71.0%       | \$−0.71        | −2.2%        |
| 3.5          | 32         | 17        | 53.1%        | 52.8%       | \$+0.12        | +0.4%        |
| 4.5          | 32         | 10        | 31.2%        | 32.5%       | \$−0.40        | −1.2%        |
| 5.5          | 32         | 6         | 18.8%        | 20.2%       | \$−0.45        | −1.4%        |
| 6.5          | 14         | 1         | 7.1%         | 9.0%        | \$−0.26        | −1.9%        |
| <b>Total</b> | <b>142</b> | <b>56</b> | <b>39.4%</b> | –           | <b>\$−1.70</b> | <b>−1.2%</b> |

Table 4: Totals backtest by actual match outcome (all scores known).

| Actual result type                                                | Bets | Net PnL | ROI   |
|-------------------------------------------------------------------|------|---------|-------|
| Strong team wins asymmetrically ( $\geq 3$ goals, weak $\leq 1$ ) | 93   | \$+0.38 | +0.4% |
| Strong team wins, but not asymmetrically                          | 29   | \$−0.84 | −2.9% |
| Draw                                                              | 20   | \$−1.24 | −6.2% |
| Upset (weak team wins)                                            | 0    | –       | –     |

### 3.4 Empirical validation of the underlying signal

Even though the total-goals market absorbs most of the asymmetry signal, the underlying hypothesis is strongly confirmed in the raw outcomes:

- The weaker team (implied win prob  $< 35\%$ ) scored **zero goals in 50% of matches** (16/32).
- The weaker team scored **zero or one goal in 97% of matches** (31/32).

If individual team scoring markets existed at reasonable prices (e.g. “team B scores under 1.5 goals”), this would represent a robust edge at this favouritism threshold.

### 3.5 Discussion

The over-3.5 line is essentially flat (+0.4%), and the over-2.5 line is slightly negative (−2.2%), indicating the Kalshi total-goals market is near-efficiently priced for strong-favourite matches. The hypothesis partially holds for the *asymmetric subset* (+0.4% ROI), but three low-scoring heavy-favourite upsets – Spain 0–0 Cape Verde (93% favourite), Ecuador 0–0 Curaçao (85%), and Germany 1–1 Paraguay (71%) – dominate the loss side and flatten the edge.

The absence of any upset (weak team winning) across all 32 matches corroborates the market’s win-probability pricing, while the rare low-scoring draw remains the primary risk to the asymmetric hypothesis.

Table 5: Combined backtest summary.

| Backtest                      | Matches | Bets | Net PnL | ROI   |
|-------------------------------|---------|------|---------|-------|
| WDL moneyline value betting   | 72      | 182  | \$-4.15 | -2.3% |
| of which: SELL implied > 80%  | –       | 17   | \$+0.96 | +5.6% |
| Total-goals BUY (fav > 65%)   | 32      | 142  | \$-1.70 | -1.2% |
| of which: asymmetric mmatches | –       | 93   | \$+0.38 | +0.4% |

## 4 Summary

Neither strategy generates significant profit after accounting for the vig embedded in Kalshi’s implied probabilities. The most actionable takeaways are:

1. Selling overpriced moneyline contracts where market implied > 80% yields a modest but consistent edge (+5.6% ROI,  $n = 17$ ).
2. The asymmetric-result signal (97% of weak teams scored  $\leq 1$  goal when the favourite implied win prob exceeded 65%) is real, but requires individual team scoring markets to exploit directly.
3. The McHale–Copula model underestimates draw probability for mismatched pairs, which is the primary source of BUY losses in asymmetric contexts.